Applied Nanofabrication for X-ray Grating Spectroscopy

Ph.D. Dissertation Defense

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December 18, 2020

Refereed Journal Articles

1. R. C. McCurdy, D. M. Miles, **Jake A. McCoy**, *et al.*, "Diffraction efficiency of a small-period astronomical x-ray reflection grating fabricated using thermally activated selective topography equilibration", *Journal of Astronomical Telescopes, Instruments, and Systems*, vol. 6, no. 4, pp. 1–19, Nov. 2020
2. **Jake A. McCoy**, M. A. Verschuuren, D. M. Miles, *et al.*, "X-ray verification of sol-gel resist shrinkage in substrate-conformal imprint lithography for a replicated blazed reflection grating", *OSA Continuum*, vol. 3, no. 11, pp. 3141–3156, Nov. 2020
3. **Jake A. McCoy**, R. L. McEntaffer, and D. M. Miles, "Extreme ultraviolet and soft x-ray diffraction efficiency of a blazed reflection grating fabricated by thermally activated selective topography equilibration", *ApJ*, vol. 891, no. 2, p. 114, Mar. 2020
4. T. Rogers, R. McEntaffer, **J. McCoy**, *et al.*, "Induced X-ray fluorescence background for high-voltage space based detectors", *Experimental Astronomy*, vol. 49, no. 2, pp. 1–20, Jan. 2020
5. D. M. Miles, S. V. Hull, T. B. Schultz, *et al.*, "Water Recovery X-Ray Rocket grating spectrometer", *Journal of Astronomical Telescopes, Instruments, and Systems*, vol. 5, no. 4, p. 044 006, 2019
6. D. M. Miles, **Jake A. McCoy**, R. L. McEntaffer, *et al.*, "Fabrication and diffraction efficiency of a large-format, replicated x-ray reflection grating", *ApJ*, vol. 869, no. 2, p. 95, Dec. 2018
7. **Jake A. McCoy**, R. L. McEntaffer, and C. Eichfeld, "Fabrication of astronomical x-ray reflection gratings using thermally activated selective topography equilibration", *Journal of Vacuum Science and Technology B: Nanotechnology and Microelectronics*, vol. 36, no. 6, Nov. 2018
8. T. Rogers, R. McEntaffer, T. Schultz, *et al.*, "Gaseous electron multiplier gain characteristics using low-pressure Ar/CO2", *Experimental Astronomy*, vol. 43, no. 2, pp. 201–210, Apr. 2017
9. **Jake McCoy**, T. Schultz, J. Tutt, *et al.*, "A Primer for Telemetry Interfacing in Accordance with NASA Standards Using Low Cost FPGAs", *Journal of Astronomical Instrumentation*, vol. 5, no. 1, 1640002, p. 1 640 002, Dec. 2016

Conference Proceedings

1. R. C. McCurdy, R. L. McEntaffer, **Jake A. McCoy**, *et al.*, "Fabrication and diffraction efficiency of a 160-nm period x-ray reflection grating produced using thermally activated selective topography equilibration", in *Optics for EUV, X-Ray, and Gamma-Ray Astronomy IX*, S. L. O'Dell and G. Pareschi, Eds., International Society for Optics and Photonics, vol. 11119, SPIE, 2019, pp. 302–310
2. D. M. Miles, R. L. McEntaffer, B. D. Donovan, *et al.*, "Grating design for the Water Recovery X-ray Rocket", in *Space Telescopes and Instrumentation 2018: Ultraviolet to Gamma Ray*, J.-W. A. den Herder, S. Nikzad, and K. Nakazawa, Eds., International Society for Optics and Photonics, vol. 10699, SPIE, 2018, pp. 1558–1565
3. M. A. Verschuuren, **Jake McCoy**, R. P. Huber, *et al.*, "AutoSCIL 200mm tooling in production, x-ray optics, and cell growth templates", in *Novel Patterning Technologies 2018*, E. M. Panning, Ed., International Society for Optics and Photonics, vol. 10584, SPIE, 2018, pp. 185–197
4. D. M. Miles, R. L. McEntaffer, T. B. Schultz, *et al.*, "An introduction to the water recovery x-ray rocket", in *UV, X-Ray, and Gamma-Ray Space Instrumentation for Astronomy XX*, O. H. Siegmund, Ed., International Society for Optics and Photonics, vol. 10397, SPIE, 2017, pp. 205–212
5. **Jake McCoy**, R. McEntaffer, and C. DeRoo, "New lithographic techniques for x-ray spectroscopy", in *Space Telescopes and Instrumentation 2016: Ultraviolet to Gamma Ray*, J.-W. A. den Herder, T. Takahashi, and M. Bautz, Eds., International Society for Optics and Photonics, vol. 9905, SPIE, 2016, pp. 613–622
6. T. Rogers, T. Schultz, **J. McCoy**, *et al.*, "First results from the OGRESS sounding rocket payload", in *UV, X-Ray, and Gamma-Ray Space Instrumentation for Astronomy XIX*, O. H. Siegmund, Ed., International Society for Optics and Photonics, vol. 9601, SPIE, 2015, pp. 13–24
7. **Jake McCoy**, T. Schultz, J. Tutt, *et al.*, "A primer for telemetry interfacing in accordance with NASA standards using low cost FPGAs", in *UV, X-Ray, and Gamma-Ray Space Instrumentation for Astronomy XIX*, O. H. Siegmund, Ed., International Society for Optics and Photonics, vol. 9601, SPIE, 2015, pp. 13–24

Jargon: *thermally-activated selective topography equilibration (TASTE), substrate-conformal imprint lithography (SCIL)*
[Schleunitz, et al. (2014), Verschuuren (2010)]

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1. **Astrophysical Motivation for Custom Blazed Gratings**
   - Soft X-ray Spectroscopy of Highly-Charged Ions in Circumgalactic & Intergalactic Media
   - Technological Development of X-ray Reflection Gratings
2. Beamline Characterization of Diffraction Efficiency
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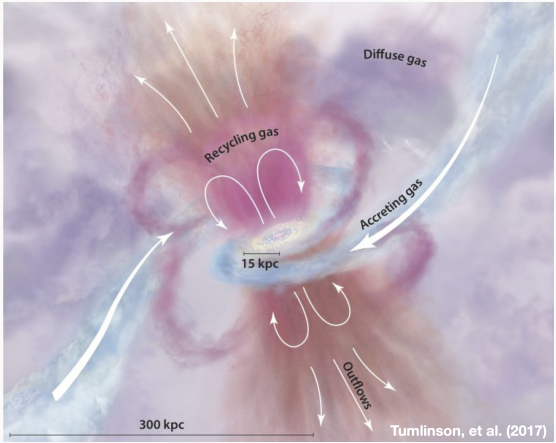
5. Conclusions & Outlook

Cosmic baryon cycle

- $\mathcal{Z} > 2$ elements (*metals*) dominated by baryons generated during life cycles of stars with $\gtrapprox 8 M_\odot$
- CGM & IGM populated with metals by material outflows from star-forming galaxies, AGN
- *Soft x-ray spectroscopy* has exclusive ability to diagnose highly-ionized portions of CGM & IGM with $T_{\rm ion} > 10^6$ K

CGM: *circumgalactic media*
(ICM: *intracluster media*)
IGM: *intergalactic media*
AGN: *active galactic nuclei*

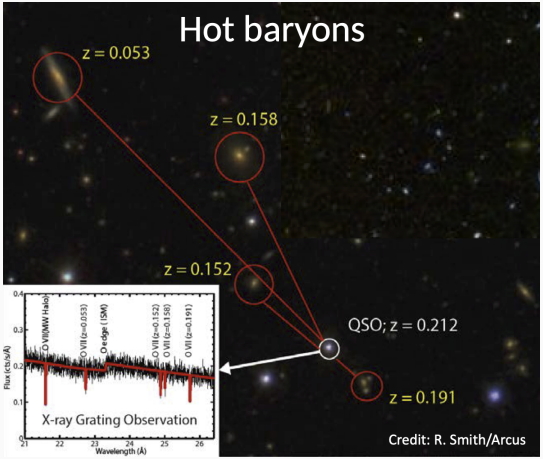


Astrophysical Motivation for Custom Blazed Gratings

	Hydrogen-like ions			Helium-like ions		
element ($\mathcal{Z}$)	ion	transition	spectral band	ion	transition	spectral band
hydrogen (1)	H I	10.2 eV	FUV/VUV	-	-	-
helium (2)	He II	40.8 eV	EUV	He II	21.2 eV	VUV/EUV
oxygen (8)	O VIII	653.5 eV	soft x-ray	O VII	573.9 eV	soft x-ray
carbon (6)	C VI	367.5 eV	soft x-ray	C V	307.9 eV	soft x-ray
neon (10)	Ne X	1021.5 eV	soft x-ray	Ne IX	922.0 eV	soft x-ray
nitrogen (7)	N VII	500.2 eV	soft x-ray	N VI	430.7 eV	soft x-ray
magnesium (12)	Mg XII	1471.7 eV	soft x-ray	Mg XI	1352.2 eV	soft x-ray
silicon (14)	Si XIV	2004.3 eV	soft x-ray	Si XIII	1865.0 eV	soft x-ray
iron (26)	Fe XXVI	6952.0 eV	x-ray	Fe XXV	6700.4 eV	x-ray
electric dipole:	$2p\ ^2P_{3/2} \leftrightarrow 1s\ ^2S_{1/2}$			$1s\,2p\ ^1P_1 \leftrightarrow 1s^2\ ^1S_0$		(data obtained from AtomDB)

- H-like & He-like ions - most extreme charge states
- Oxygen is most abundant $\mathcal{Z} > 2$ element
- Electric-dipole transitions (*resonance lines*) have largest transition rates, appear strongest in spectrum

Astrophysical Motivation for Custom Blazed Gratings



- The *emission measure*, EM $\equiv \int n_e n_{\rm ion}\, \mathrm{d}^3\boldsymbol{r}$, is very low in diffuse CGM, IGM
- Absorption spectroscopy using AGN continua beneficial for detecting weak spectral lines
- Measure spectral lines at various redshifts, *z*, correlate with intercepting galaxies

Astrophysical Motivation for Custom Blazed Gratings

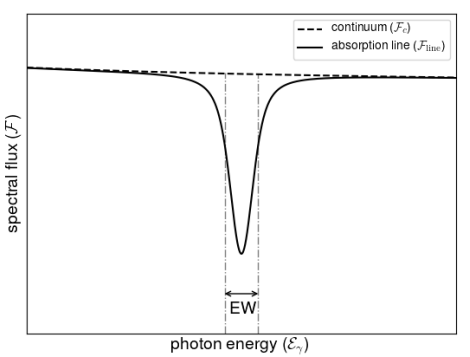


Figure: Voigt profile of absorption line, $\mathcal{F}_{\rm line}$, with underlying AGN continuum: $\mathcal{F}_c \propto \mathcal{E}_\gamma^{-0.7}$

equivalent width: $\mathrm{EW} \equiv \int \left(\frac{\mathcal{F}_c - \mathcal{F}_{\rm line}}{\mathcal{F}_c} \right) \mathrm{d}\mathcal{E}_\gamma$

Energy-dispersive spectroscopy ($\mathcal{R} = \mathcal{E}_\gamma / \Delta\mathcal{E}_\gamma$)

- unresolved line: $F_{\rm line} \equiv [\mathcal{F}_c - \mathcal{F}_{\rm line}]\, \Delta\mathcal{E}_\gamma$
- photons: $N_{\rm line} = F_{\rm line} A_{\rm col} t_{\rm obs}$ with $N_{\rm tot} = N_c - N_{\rm line}$

Equivalent width of weak, unresolved spectral line

$$\mathrm{EW} = \frac{F_{\rm line}}{\mathcal{F}_c} \approx \tau_\gamma^{\rm ion} \Delta\mathcal{E}_\gamma \;\text{ with }\; \tau_\gamma^{\rm ion} \equiv \sigma_\gamma^{\rm ion} \mathcal{N}_\gamma^{\rm ion} \;\; (\textit{optical depth})$$

- EW yields $\tau_\gamma^{\rm ion} \implies$ ion density can be inferred
- Detection depends on signal-to-noise: $\mathcal{S} = N_{\rm line}/\sqrt{N_{\rm tot}}$ with $N_{\rm tot} \approx N_c = N_{\rm line} \Delta\mathcal{E}_\gamma / \mathrm{EW}$

Instrument figure of merit for weak absorption line

$$\mathrm{FOM}_{\rm weak} \propto \frac{\mathcal{S}}{\sqrt{F_{\rm line} t_{\rm obs}}} \approx \sqrt{\frac{A_{\rm col}\mathrm{EW}}{\Delta\mathcal{E}_\gamma}} \propto \sqrt{A_{\rm col}\mathcal{R}}$$

High spectral resolving power & large collecting area both required for sensitive spectroscopy

Astrophysical Motivation for Custom Blazed Gratings

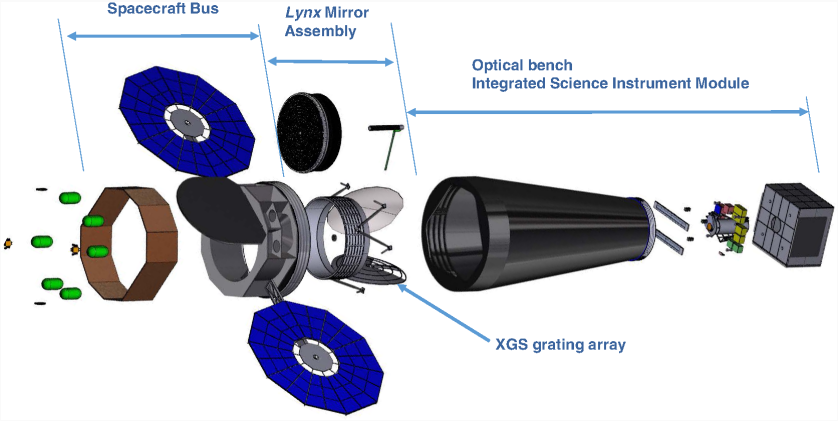


Figure: An exploded view of the *Lynx X-ray Observatory* (McEntaffer 2019)

cf. Gaskin, et al. (2019)

***Lynx* spectroscopy**

- *Lynx X-ray Microcalorimeter (LXM)*
  - $\mathcal{R} = \mathcal{E}_\gamma / \Delta\mathcal{E}_\gamma$
- *X-ray Grating Spectrometer (XGS)*
  - $\mathcal{R} = \lambda / \Delta\lambda$
- Compare *LXM* and *XGS* for spectroscopy of CGM & IGM
- *X-ray reflection gratings* & *critical-angle transmission gratings* considered for *XGS*

Astrophysical Motivation for Custom Blazed Gratings

PennState

The need for next-generation x-ray gratings

- O VIII and O VII important for absorption studies
  - High $\sqrt{A_{\rm col}\mathcal{R}}$ required for $\mathcal{E}_\gamma \gtrapprox 650\,{\rm eV}/(1+z)$
  - $\mathcal{R} \gtrapprox 5000$ desirable for resolving thermally-broadened oxygen line with $T_{\rm ion} > 10^6$ K
- *LXM* baselines $A_{\rm col} \approx 1000\,{\rm cm}^2$ and $\Delta\mathcal{E}_\gamma \approx 3\,{\rm eV}$
  - $\mathcal{R} = \mathcal{E}_\gamma/\Delta\mathcal{E}_\gamma \gtrapprox 350$ for $\mathcal{E}_\gamma \gtrapprox 1\,{\rm keV}$
- *XGS* baselines $A_{\rm col} \gtrapprox 4000\,{\rm cm}^2$ and $\Delta\lambda \approx 4\,{\rm mÅ}$
  - $\mathcal{R} = \lambda/\Delta\lambda \gtrapprox 5000$ for $n\lambda \gtrapprox 2\,{\rm nm}$
  - ${\rm FOM}_{\rm weak} \lessapprox 8$ that of *LXM*, $\lessapprox 20$ that of *XMM-Newton*

Lynx science goal

Probe density of CGM, IGM through measuring EW $\gtrapprox 1\,{\rm mÅ}$ for many AGN-absorbing systems, correlating measured z with those of intercepting galaxies

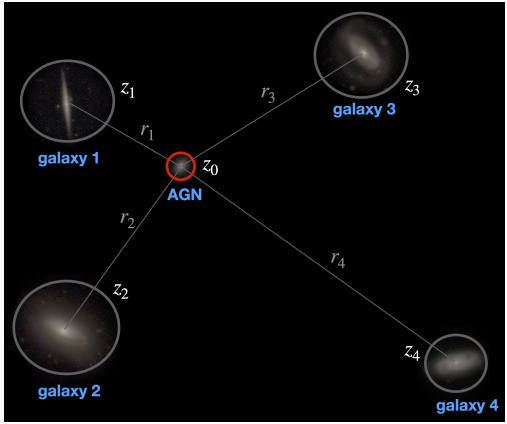


Figure: Cartoon of an AGN at a redshift z_0 with four galaxies at smaller redshifts z_i (for $i = 1$ to 4) in the foreground for absorption spectroscopy
(Depicted galaxies are optical images obtained from Galaxy Zoo)

Astrophysical Motivation for Custom Blazed Gratings

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Main technological challenges of grating fabrication

1. Producing a master grating with a sawtooth surface-relief profile that enables
   - Total absolute diffraction efficiency $> 40\,\%$
   - $\mathcal{R} \gtrapprox 5000$ in a Wolter-I telescope (variable-line-space groove layout)
2. Mass-replicating thousands of blazed gratings to populate modular grating arrays

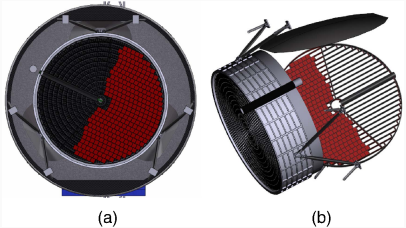

Figure: Modular reflection grating arrays (red) for *XGS* relative to *Lynx Mirror Assembly* (McEntaffer 2019)

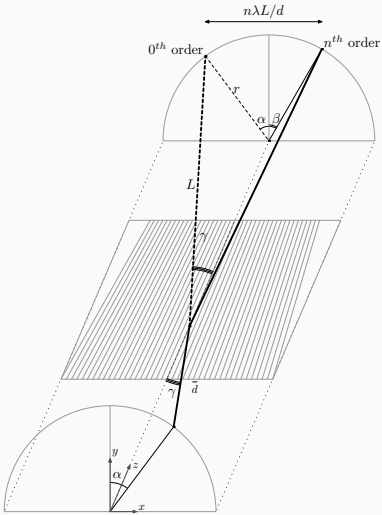

Figure: Reflection grating geometry for conical diffraction with radial groove layout (not to scale)

Astrophysical Motivation for Custom Blazed Gratings

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X-ray reflection grating fabrication

1. Mechanical ruling engine
   - Diamond-tip burnishing
   - 182 replicas cast in resin for *XMM-Newton RGS*
2. Holographic recording
   - Generate sinusoid in *photoresist* following wet development
   - Produces parallel or curved groove layout
3. Electron-beam lithography (EBL)
   - Pattern laminar structure in *polymeric resist* using high-energy electrons, wet development
   - Couple with *crystallographic etching* to realize sawtooth
   - ☐ Franke, et al. (1997), McEntaffer et al. (2013), Miles, et al. (2018)
4. Nanoimprint lithography (NIL)
   - Use silicon master to imprint replicas in resist
   - NANONEX UV-NIL pursued previously
   - ☐ Chang (2003,2004), McEntaffer et al. (2013), Miles, et al. (2018)

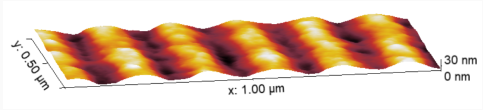


Figure: *Atomic force micrograph (AFM)* of sinusoidal grating used for sounding rockets (JOBIN YVON)

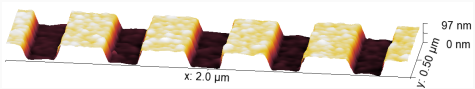


Figure: AFM of laminar grating dry etched in amorphous silicon (Penn State Nanofab)

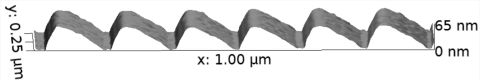


Figure: AFM of blazed grating fabricated by UV-NIL from KOH-etched master (Image credit: T. Tighe)

Astrophysical Motivation for Custom Blazed Gratings

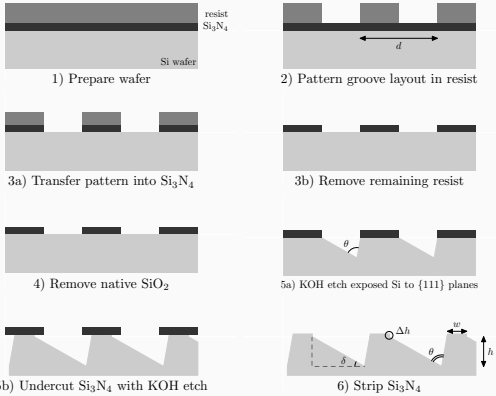


Figure: Outline of fabrication recipe for crystallographic etching in ⟨311⟩-oriented silicon wafer using EBL

cf. Miles, McCoy, McEntaffer, et al. (2018)

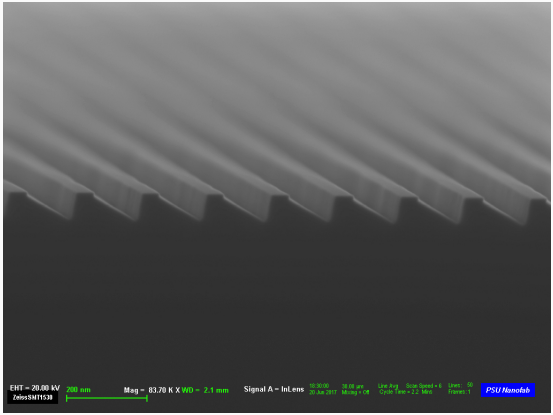


Figure: Field emission scanning electron micrograph (FESEM) of $d \lessapprox 160$ nm KOH-etched grating

Astrophysical Motivation for Custom Blazed Gratings

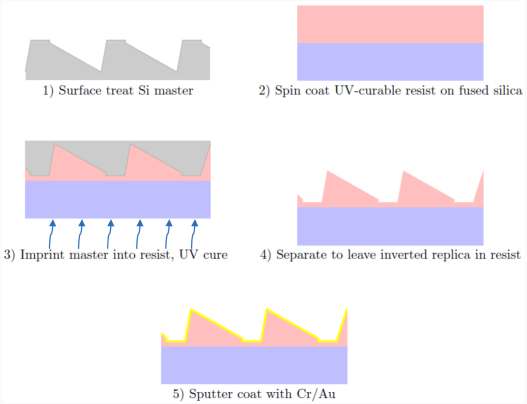


Figure: Outline of UV-NIL grating replication process using KOH-etched as stamp

cf. Miles, et al. (2018)

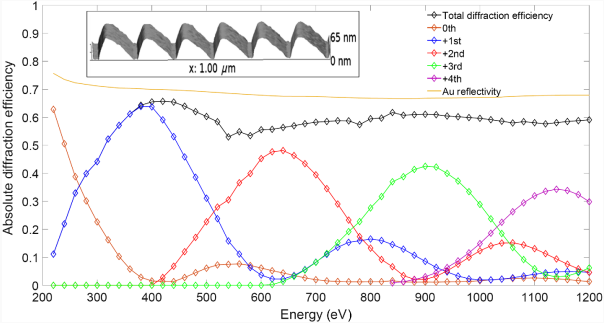
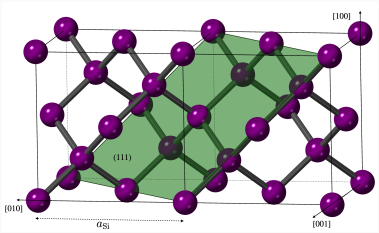


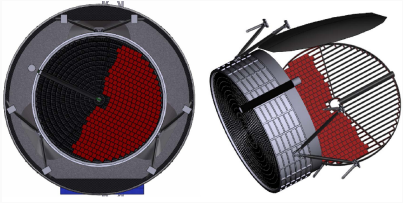
Figure: Measured diffraction efficiency of gold-coated UV-NIL imprint meets *Lynx* sensitivity requirements (Miles, et al. 2018)

Lynx XGS requirement

Total diffraction efficiency $>$ 40 % in the soft x-ray

Astrophysical Motivation for Custom Blazed Gratings





Images: T. Wood using CrystalMaker (*top*), McEntaffer 2019 (*bottom*)

Grating fabrication limitations

1. Dependence on silicon crystal structure
   - Cubic structure of silicon prevents formation of fanned grooves with smooth, continuous triangular facets

   $\Longrightarrow$ Limits $\mathcal{R}$ and hence $\mathrm{FOM}_{\mathrm{weak}} \propto \sqrt{A_{\mathrm{col}}\mathcal{R}}$
2. Throughput of UV-NIL
   - Rigidity of thick silicon stamp requires applied pressure to achieve conformal contact
   - High pressure can lead to imperfections from particulate contaminants, trapped air pockets
   - Tens of quality imprints from single stamp

   $\Longrightarrow$ Impractical for *XGS*, sounding rockets

1 Astrophysical Motivation for Custom Blazed Gratings

2 **Beamline Characterization of Diffraction Efficiency**

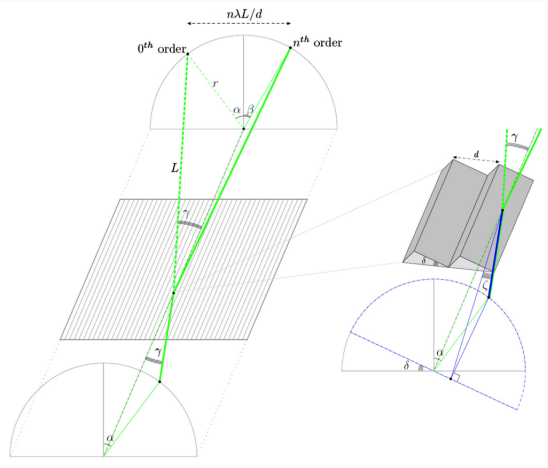
- Grating Testing at the Advanced Light Source (ALS)
- Theoretical Modeling using PCGRATE-SX

3 Grating Fabrication by Thermally-Activated Selective Topography Equilibration (TASTE)

4 Grating Replication by Substrate-Conformal Imprint Lithography (SCIL)

5 Conclusions & Outlook

Beamline Characterization of Diffraction Efficiency



$$\sin(\alpha) + \sin(\beta) = \frac{n\lambda}{d\sin(\gamma)} \quad \text{for } n = 0, \pm1, \pm2, \pm3...$$

X-ray grating considerations

1. $250\,\text{eV} \lessapprox \hbar\omega \lessapprox 2\,\text{keV}$ coincides with K-shell binding energies in low-to-mid $\mathcal{Z}$ atoms
   - Soft x-rays easily absorbed by air
   - High-$\mathcal{Z}$ materials beneficial for reflectivity
2. $5\,\text{nm} \gtrapprox \lambda \gtrapprox 0.5\,\text{nm}$ approaches atomic scale
   - Coherent scattering occurs small angles
   - Critical angle for *total external reflection*: $\zeta_c(\omega)$
3. Extreme off-plane geometry
   - $\sin(\zeta) = \sin(\gamma)\cos(\delta - \alpha)$ with $\zeta < \zeta_c(\omega)$
   - Tune *d* and δ to maximize diffraction efficiency near λ_b with $\beta = 2\delta - \alpha$

Blaze wavelength

$$\lambda_b = \frac{d\sin(\gamma)}{n}\left[\sin(\alpha) + \sin(2\delta - \alpha)\right]$$

Littrow configuration: $\alpha = \beta = \delta$ with $\zeta = \gamma$

Beamline Characterization of Diffraction Efficiency

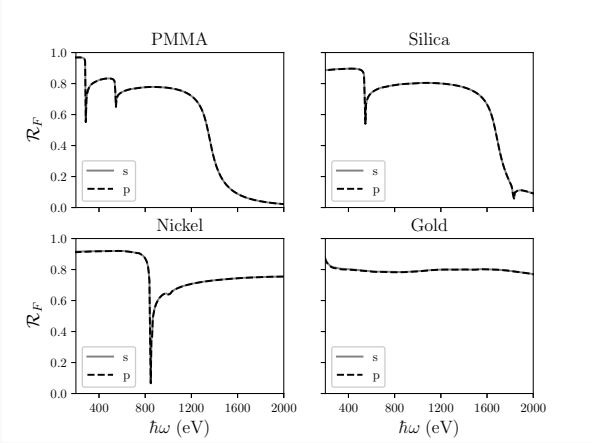


Figure: Fresnel soft x-ray reflectivity at $\zeta = 1°$ (CXRO)

$$\mathcal{R}_F \approx \left\| \frac{\sin(\zeta) - \sqrt{\tilde{\nu}^2(\omega) - \cos^2(\zeta)}}{\sin(\zeta) + \sqrt{\tilde{\nu}^2(\omega) - \cos^2(\zeta)}} \right\|^2$$

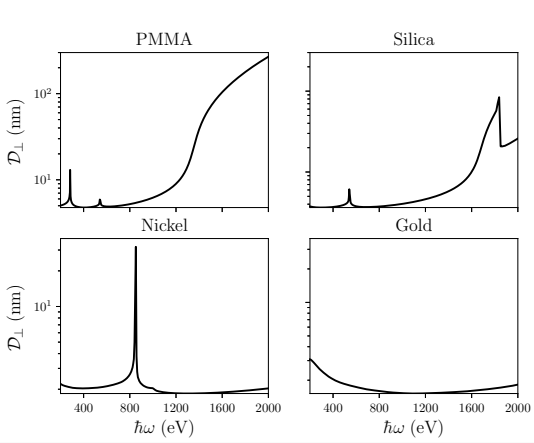


Figure: Total external reflection penetration depth at $\zeta = 1°$

$$\mathcal{D}_\perp = \frac{\lambda}{4\pi\ \mathrm{Im}\left[\sqrt{\tilde{\nu}^2(\omega) - \cos^2(\zeta)}\right]}$$

Beamline Characterization of Diffraction Efficiency

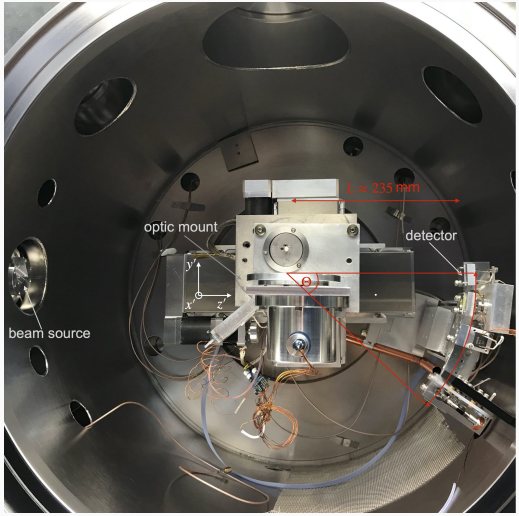


Figure: Overview of beamline 6.3.2 of the ALS

Advanced Light Source (ALS) beamline 6.3.2

- Station for EUV and soft x-ray reflectometry
  - Highly-coherent, monochromatic beam travels along the z'-direction to photodiode
  - Detector staging moves linearly in x'-direction, goniometrically in Θ-direction
- Use staging translational degrees of freedom (DOF) to position grating
- Use rotational DOF to set grating geometry in grazing-incidence, near-Littrow configuration
- Determine *absolute diffraction efficiency*, $\mathcal{E}_n$, by measuring intensity of each propagating order, $\mathcal{I}_n$, to unobstructed beam, $\mathcal{I}_{\text{inc}}$

Absolute diffraction efficiency

$$\mathcal{E}_n \equiv \frac{\mathcal{I}_n}{\mathcal{I}_{\text{inc}}} \quad \text{for } n = 0, \pm 1, \pm 2, \pm 3...$$

Beamline Characterization of Diffraction Efficiency

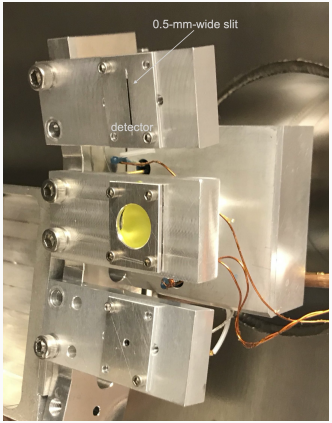
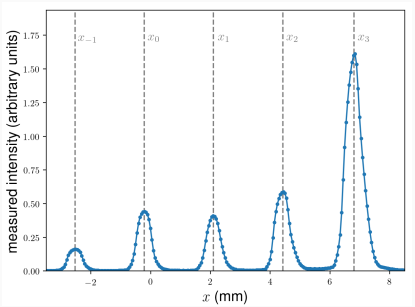


Figure: View of masked photodiode detector on goniometric staging



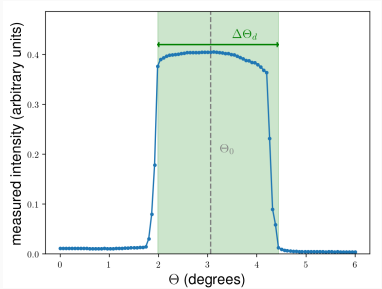
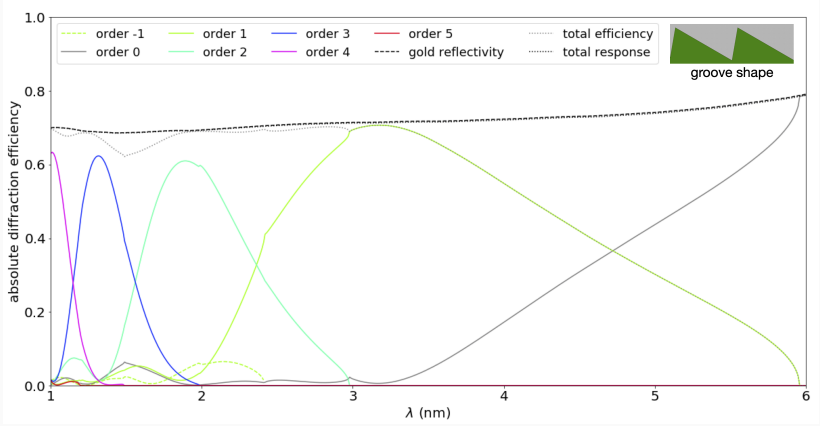


Figure: Example of measured intensity data for diffracted arc

1. Equip detector with 0.5 mm-wide, vertical slit $\Longrightarrow$ order separation
   - $\Delta x_n = n\lambda L/d \sim$ millimeters with $L \approx \ell_{\text{det}}/\Delta\Theta_d \approx 235\,\text{mm}$
2. Gather (x_n, y_n) coordinates of propagating orders $\Longrightarrow$ diffracted arc
   - $y_n = L\sin(\Theta_n)$ using goniometric staging
3. Determine γ and α from arc radius, relative position of direct beam
   - Adjust principal-axis angles for desired geometry

Modeling diffraction efficiency

- PCGRATE-SX solves Helmholtz equation via *integral method* for custom grating boundary, γ and α
- Take into account reflectivity of groove facet
- Efficiency response for x-ray reflection gratings polarization insensitive (Marlowe, et al. 2016)



1 Astrophysical Motivation for Custom Blazed Gratings

2 Beamline Characterization of Diffraction Efficiency

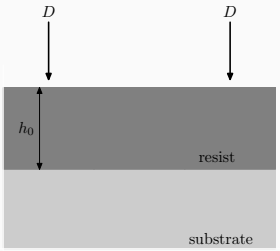
3 **Grating Fabrication by Thermally-Activated Selective Topography Equilibration (TASTE)**

- Process Development for TASTE & Grating Prototype Fabrication
- Beamline Experiments
- Analysis & Discussion for Diffraction Efficiency

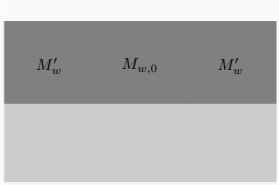
4 Grating Replication by Substrate-Conformal Imprint Lithography (SCIL)

5 Conclusions & Outlook

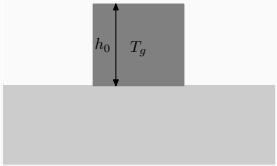
Grating Fabrication by TASTE



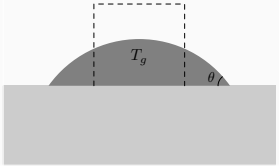
1) Electron exposure with a fixed dose D



2) Dose causes M_w to be reduced from $M_{w,0}$ to M'_w



3) Wet development leaves laminar structure with T_g



4) Patterned resist can be reflowed at $T > T_g$

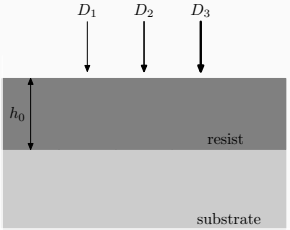
Figure: Outline of a traditional thermal reflow process

EBL and thermal reflow

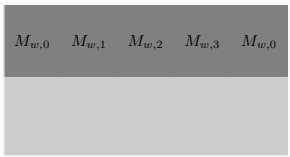
- Electron exposure with *dose D* reduces *molecular weight* M_w via chain scission
- Exposed resist is soluble in developer, clear to substrate
- Remaining resist has *glass-liquid transition temperature* T_g
  - Depends on M_w
  - 100 °C $\lessapprox T_g \lessapprox$ 130 °C for PMMA
- $T \gtrapprox T_g + 50$ °C thermal treatment causes molten resist to equilibrate slowly into energetically-favorable, convex topography

cf. Kirchner, et al. (2014)

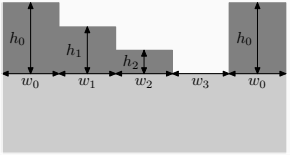
Grating Fabrication by TASTE



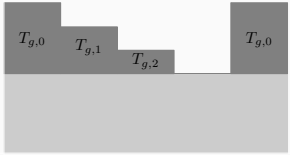
1) Dose-modulated electron exposure by GEBL



2) A lateral gradient in M_w is produced



3a) Wet development produces stepped structure



3b) Patterned resist features lateral gradient in T_g

Figure: Properties of GEBL-processed resist processed that enable *thermally-activated selective topography equilibration (TASTE)*

Grayscale EBL (GEBL)

- Dose-modulated electron exposure imparts lateral M_w gradient
  - $\Longrightarrow$ $D_1 < D_2 < D_3$ yields $M_{w,1} > M_{w,2} > M_{w,3}$
- M_w-dependent etch rates give rise to stepped structure
  - $\Longrightarrow$ $M_{w,1} > M_{w,2} > M_{w,3}$ yields $h_1 > h_2 > h_3 = 0$ (timed wet development)
- GEBL structure exhibits lateral gradient in T_g
  - $\Longrightarrow$ $M_{w,0} > M_{w,1} > M_{w,2}$ yields $T_{g,0} > T_{g,1} > T_{g,2}$
  - $\Longrightarrow$ **Enables selective thermal reflow** at $T_{g,0} > T > T_{g,1}$
- Need experimentally-determined *resist-contrast curve*: $h(D)$

cf. Schleunitz, et al. (2014)

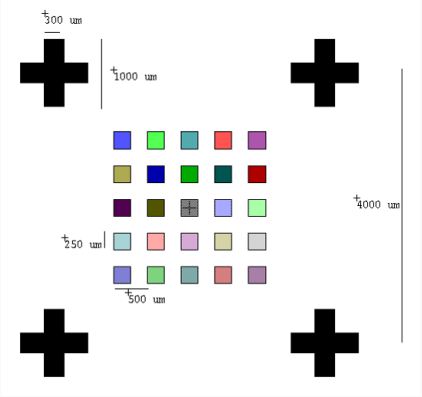


Figure: CAD (TANNER L-EDIT) for resist-contrast data collection

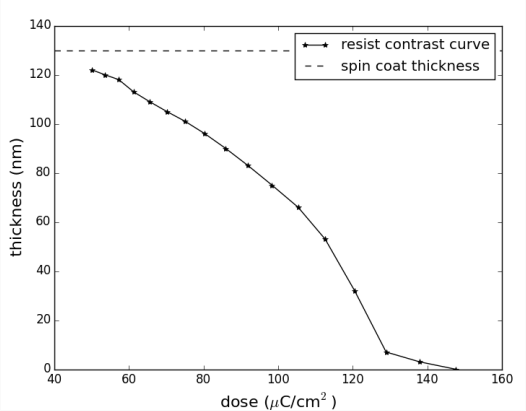


Figure: Resist contrast for 130 nm-thick, 950 kg mol^{-1} PMMA using 1:1 MIBK/IPA for 2 min, IPA for 30 s

McCoy, et al. (2018)

Grating Fabrication by TASTE

PennState

GEBL process development

- Input resist-contrast data to BEAMER 3DPEC algorithm for *proximity-effect correction (PEC)*
- Expose and develop test patterns for GEBL

Test patterns

- *pattern A*: 6-level staircase where each step is 140 nm wide $\implies$ $d = 840$ nm
- *pattern B*: 4-level staircase where each step is 100 nm wide $\implies$ $d = 400$ nm
- *pattern C*: 4-level staircase where top step is reduced to 40 nm; remaining levels are 120 nm wide $\implies$ $d = 400$ nm

Carry out thermal reflow experimentation

- Explore *T-t* parameter space

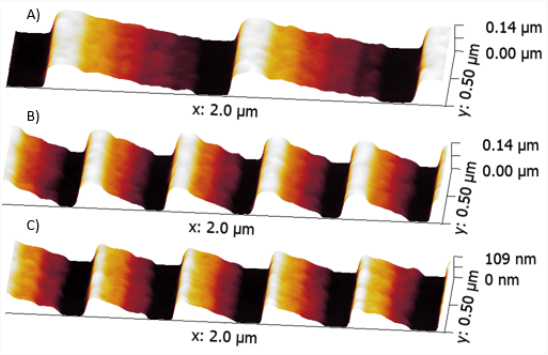


Figure: AFMs of GEBL test patterns in 130 nm-thick PMMA

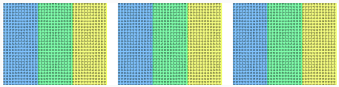


Figure: Layout BEAMER 3DPEC for GEBL

$d = 840\,\text{nm}, \delta \approx 10^\circ$

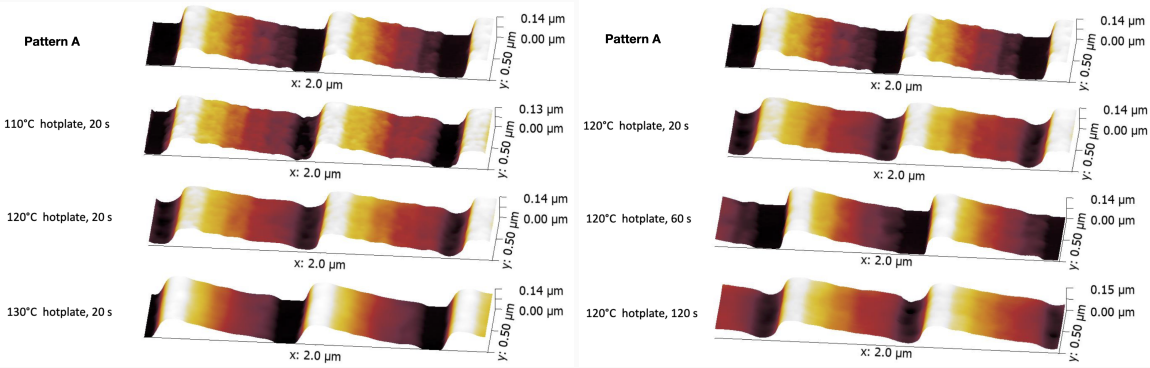


Figure: Thermal reflow results for pattern A as a function of *T* (*left*), as a function of *t* (*right*)

McCoy, et al. (2018)

Grating Fabrication by TASTE

$d = 400\,\text{nm}, \delta \approx 27^\circ$

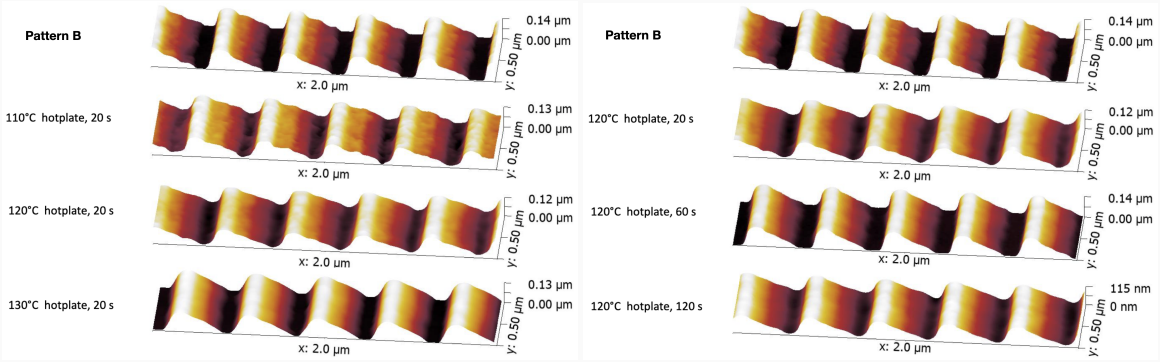


Figure: Thermal reflow results for pattern B as a function of T (*left*), as a function of t (*right*)

McCoy, et al. (2018)

PennState

Grating prototype

- Start from pattern B with $d = 400$ nm
- Design for $\alpha = \beta = \delta \approx 27°$ with $\zeta = \gamma \approx 1.7°$
- Pattern 50 mm along groove direction and 7.5 mm along dispersion direction to encompass soft x-ray beam

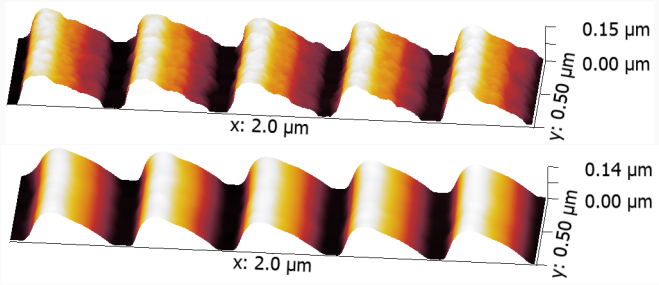


Figure: AFM of GEBL-processed resist (*top*), resist following thermal reflow at 116 °C for 30 min (*bottom*)

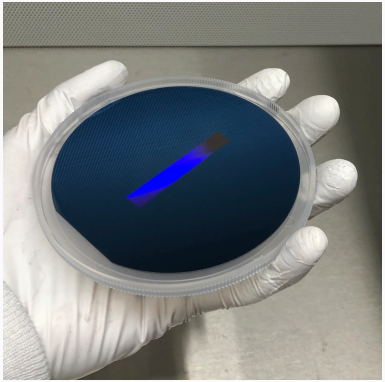


Figure: Surface-relief mold for grating prototype patterned in 130 nm-thick PMMA on 100 mm-diameter silicon wafer

McCoy, et al. (2020a)

Grating Fabrication by TASTE

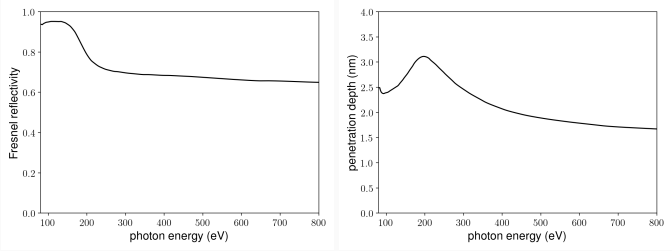


Figure: $\mathcal{R}_F$ and $\mathcal{D}_\perp$ for thick slab of gold for nominal test geometry

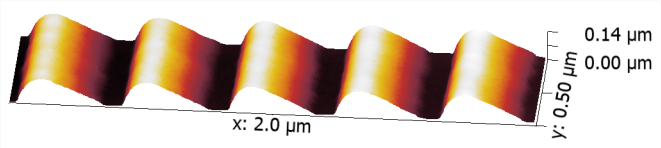


Figure: AFM of prototype grooves following EBPVD of gold using titanium as adhesion layer on PMMA

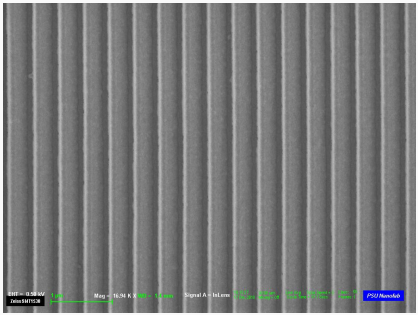


Figure: FESEM of gold-coated grooves

Coating for reflectivity

Use *electron-beam physical vapor deposition (EBPVD)* to provide 15 nm Au for *thick slab*, 5 nm Ti for adhesion

McCoy, et al. (2020a)

Grating Fabrication by TASTE

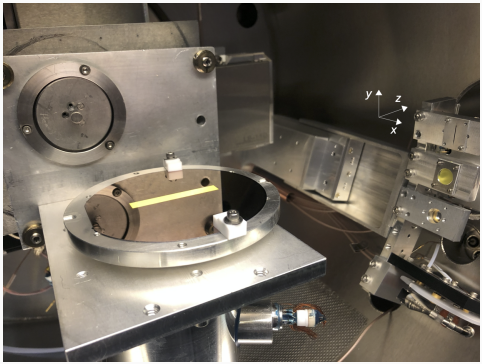


Figure: Grating prototype installed inside test chamber of ALS beamline 6.3.2

- Establish near-Littrow configuration
- Measure $\mathcal{E}_n$ across soft x-ray range with separable orders, into EUV

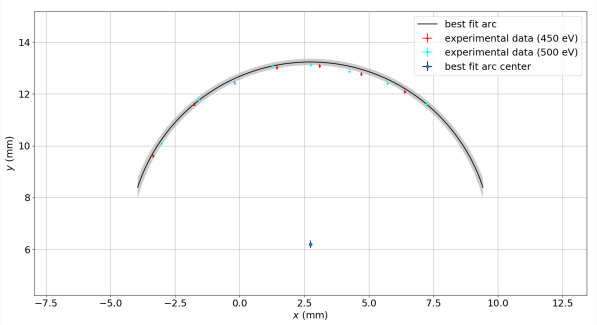


Figure: Test-configuration diffracted arc mapped using 450 eV and 500 eV data fit to circle

parameter	measured value
system throw (L)	232.0 ± 1.4 mm
arc radius (r)	7.03 ± 0.12 mm
cone opening half-angle (γ)	$1.73 \pm 0.03^\circ$
azimuthal incidence angle (α)	$23.4 \pm 0.6^\circ$

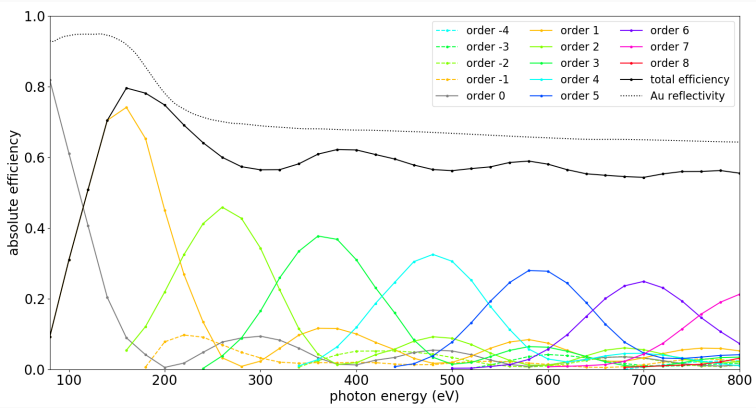


Figure: Measured absolute $\mathcal{E}_n$ for prototype compared to $\mathcal{R}_F$ of gold

$$\mathcal{E}_{\text{tot}} \equiv \sum_n \mathcal{E}_n \text{ for } n \neq 0 \implies \text{exceeds } 40\,\%$$

Measuring efficiency

- $\mathcal{E}_n = \mathcal{I}_n / \mathcal{I}_{\text{inc}}$ with dark current, diffuse scatter subtracted
  - ☐ Measure in 20 eV steps
- Peak-order $\mathcal{E}_n$ ranges from 75 % to 25 % as $\mathcal{E}_\gamma$ increases
  - ☐ Diffracted orders become suppressed with increasing n

  $\implies$ Likely due to imperfect sawtooth topography generated by TASTE
- Total efficiency in soft x-ray meets *Lynx* requirements
  - ☐ $\mathcal{E}_{\text{tot}} \approx 60\,\%$
  - ☐ $\mathcal{E}_{\text{tot}} / \mathcal{R}_F \lessapprox 95\,\%$

McCoy, et al. (2020a)

Grating Fabrication by TASTE

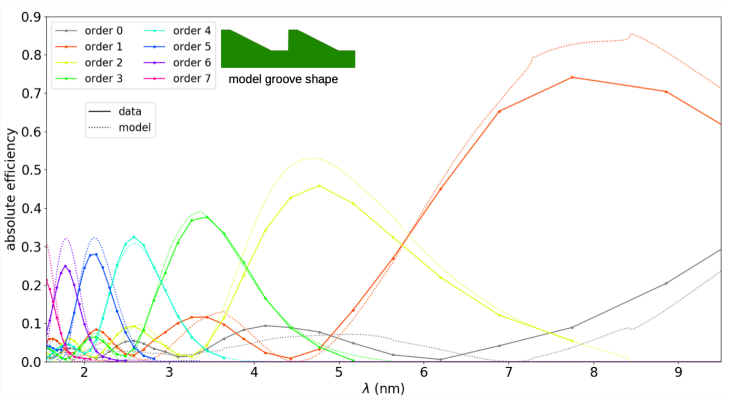


Figure: Measured $\mathcal{E}_n$ compared to theoretical $\mathcal{E}_n$ modeled using PCGRATE-SX

Efficiency modeling

- Investigate impact of unpointed apex on measured $\mathcal{E}_n$
- Approximate groove shape as acute trapezoid with $\delta \approx 27^\circ$
  - Include flat bottom to represent cleared portion
- Keep $\gamma = 1.73^\circ$, $\alpha = 23.4^\circ$ and $d = 400$ nm fixed, vary model
- Amplitudes of peak orders generally fall short of model
- Deviations from ideal acute trapezoid expected to have impact on $\mathcal{E}_n$ distribution among orders

McCoy, et al. (2020a)

1 Astrophysical Motivation for Custom Blazed Gratings

2 Beamline Characterization of Diffraction Efficiency

3 Grating Fabrication by Thermally-Activated Selective Topography Equilibration (TASTE)

4 **Grating Replication by Substrate-Conformal Imprint Lithography (SCIL)**

- SCIL Process for X-ray Reflection Gratings
- Beamline Experiments
- Analysis & Discussion for Resist Shrinkage

5 Conclusions & Outlook

Grating Replication by SCIL

Substrate-Conformal Imprint Lithography (SCIL)

- PHILIPS SCIL NANOIMPRINT SOLUTIONS in Eindhoven, Netherlands
- Form flexible, composite stamp from rigid master template
- Use stamp (rather than master) to produce replicas in various resists

Alternative to UV-NIL

- Stamp features carried in modified PDMS
- Conforms locally to particulate contaminants, globally to bow of replica substrate
- Wave-like sequential imprinting serves to eliminate large trapped air pockets

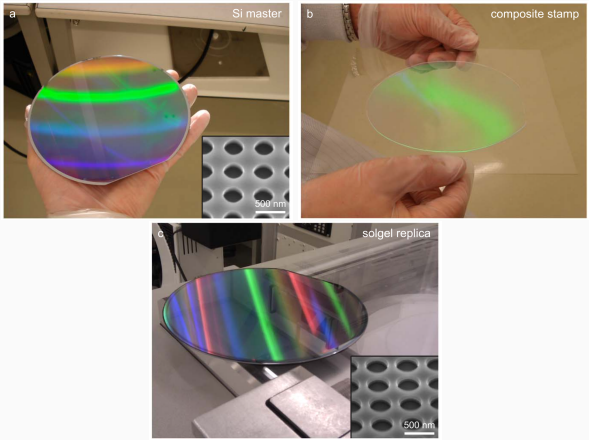


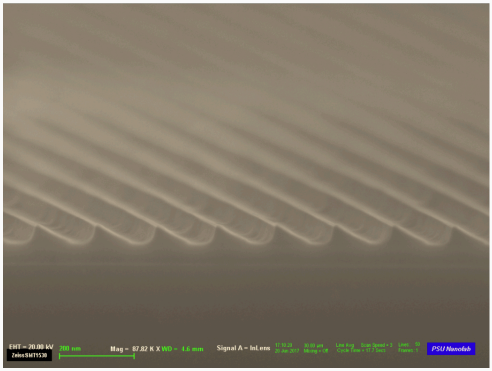
Figure: Overview of SCIL: a) master template, b) composite stamp, c) imprint in resist (Verschuuren, et al. 2019)

NanoGlass sol-gel resist

- High-volume production relies on long stamp lifetime
  - Best suited for inorganic resist that cures through silica *sol-gel* process (NanoGlass)
- Single stamp produces $\gtrapprox$ 700 imprints in NanoGlass
  - 60 wafers per hour, over 150 mm
  - Beneficial for *Lynx XGS*, sounding rockets
- AutoSCIL, NanoGlass used for *WRXR*
  - 26 replicas of 110 cm^2 master grating
- Resist shrinkage $\implies$ topographic distortion
  - Imprinted resist initially has $\sim$ 70 % density of fused silica
  - $\sim$ 10 % volume shrinkage expected for $T_{cure} \approx 90\ ^\circ C$

$\implies$ Study impact of resist shrinkage on blaze angle

Figure: AutoSCIL tool (Verschuuren, et al. 2018)



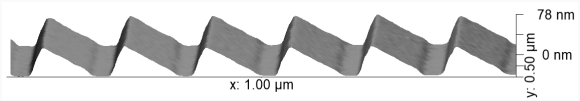


Figure: Cross-section FESEM (*top*), AFM (*bottom*) of grating imprint in NanoGlass T1100

SCIL for x-ray reflection gratings

- Use KOH-etched silicon master (Miles, et al. 2018)
  - ☐ $\delta = 30.0 \pm 0.8°$ under AFM
  - ☐ Surface-treat for anti-stiction
- Use inverted stamp to produce replicas that emulate UV-NIL gratings using $T_{\text{cure}} \approx 90\,°\text{C}$
  - ☐ $\delta' = 27.9 \pm 0.7°$
  - ☐ $\sigma \approx 0.6$ nm RMS
- Sputter-coat with 15 nm of Au, using 5 nm Cr for adhesion (Miles, et al. 2018)
  - ☐ $\delta' = 28.4 \pm 0.8°$
  - ☐ $\sigma \approx 0.8$ nm RMS

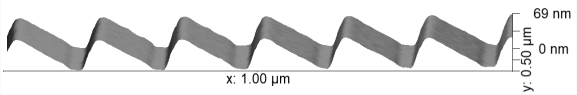


Figure: AFM of gold-coated grating imprint

McCoy, et al. (2020b)

Grating Replication by SCIL

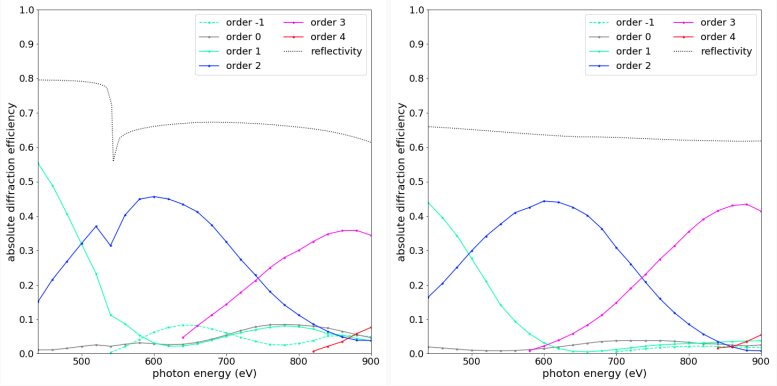


Figure: Measured absolute diffraction efficiency data for silicon master (*left*), coated SCIL replica (*right*)

$\Longrightarrow$ Comparable to UV-NIL results (Miles, et al. 2018)

Beamline experiments

- Establish near-Littrow configurations at ALS beamline 6.3.2
- Measure $\mathcal{E}_n$ for $n = 2, 3$ and compare centroids to PCGRATE-SX models
  - Most sensitive to γ and δ for fixed d

Blaze wavelength

$$\lambda_b \approx \frac{2d\gamma \sin(\delta)}{n}\left(1 - \frac{(\delta-\alpha)^2}{2}\right)$$

- Constrain δ'/δ to examine impact of resist shrinkage

McCoy, et al. (2020b)

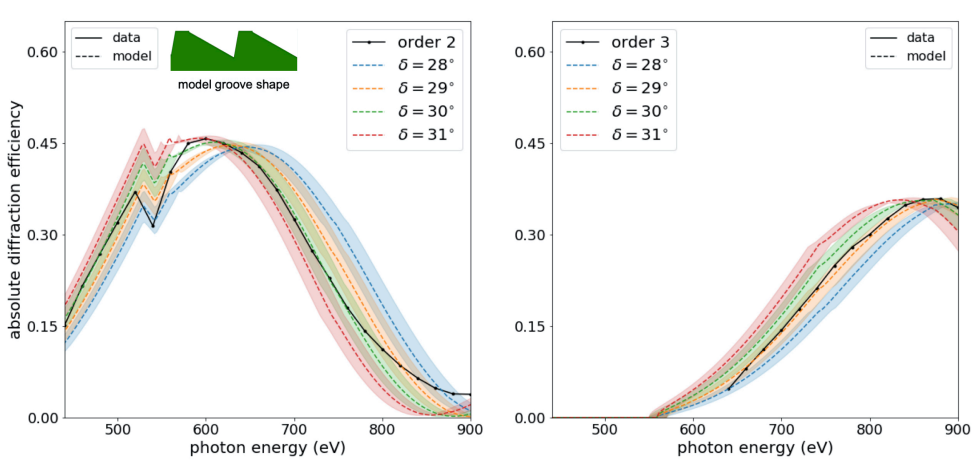


Figure: Measured $\mathcal{E}_n$ in $n = 2, 3$ for silicon master compared to PCGRATE-SX models of varying blaze angle

$\Longrightarrow$ Silicon master close to the nominal value of $\delta = 29.5°$

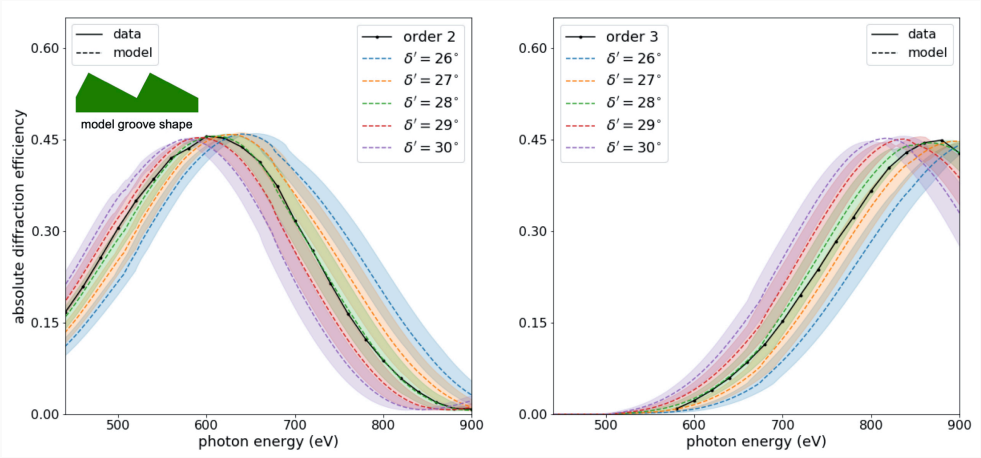


Figure: Measured $\mathcal{E}_n$ in $n = 2, 3$ for the silicon master compared to PCGRATE-SX models of varying blaze angle

$\implies$ Replica closely matches grating with $\delta' = 28°$

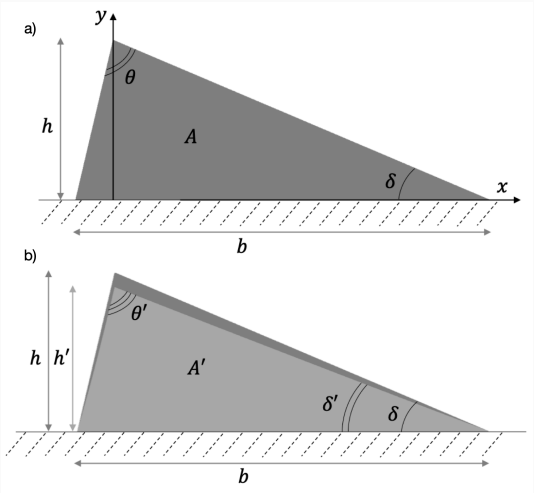


Figure: Approximate model for resist shrinkage with $y = 0$ as fixed boundary defined by residual layer

Resist shrinkage

- Assume inversion of silicon master before shrinkage
  - Facet base: $b \approx d - w \lessapprox 130\,\text{nm}$
  - Apex angle: $\theta \approx 70.5°$ defined by $\{111\}$ planes
  - Facet angles δ and $180° - \theta - \delta$
- Shrinkage w/o boundary: $V' = V(1 - \chi)$
  - Residual layer constrained laterally, serves as boundary (Horiba, et al. 2012)
  - Groove length also constrains shrinkage $\implies A' = A(1 - \chi)$ throughout cross-section
- Shrinkage aspect ratio varies with distance
  - Assume $S \equiv s_x/s_y = 1 - \mathrm{e}^{-y/\ell_e}$
- Divide into 1000 layers, require each to be reduced in area according to

$$(1 - s_x)\,(1 - s_y) = \frac{A'}{A} = \text{const.}$$

McCoy, et al. (2020b)

Resist shrinkage

- General result
  - Decrease in groove depth: $h' < h$
  - Increase in apex angle: $\theta' > \theta$
  - Small level of curvature for $y \lessapprox \ell_e$
  - **Decrease in blaze angle:** $\delta' < \delta$
- δ'/δ depends on χ and ℓ_e
  - $\delta'/\delta \approx 0.93$ consistent with $\chi \approx 0.1$ & $\ell_e \approx 0.05h$
  - $\sim 10\,\%$ shrinkage in sol-gel network expected
- Resist shrinkage in SCIL shifts blaze response
  - Comparable result found for UV-NIL replica from Miles et al. (2018)

$\Longrightarrow$ Compensate for effect in master grating to ensure that replicas perform as expected

McCoy, et al. (2020b)

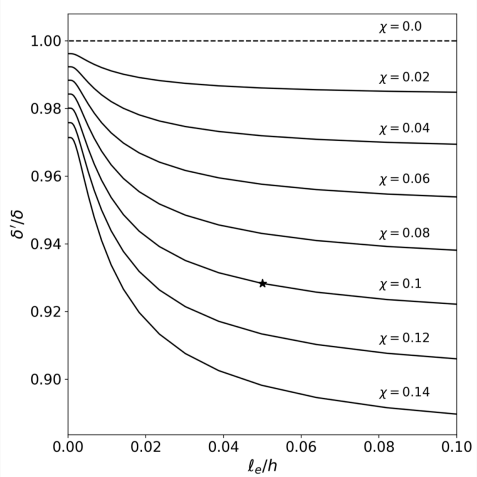


Figure: Reduced blaze angle predicted by resist-shrinkage model

1 Astrophysical Motivation for Custom Blazed Gratings

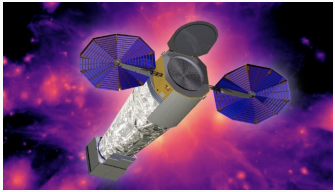
2 Beamline Characterization of Diffraction Efficiency

3 Grating Fabrication by Thermally-Activated Selective Topography Equilibration (TASTE)

4 Grating Replication by Substrate-Conformal Imprint Lithography (SCIL)

5 Conclusions & Outlook

Conclusions & Outlook



X-ray grating fabrication challenges for *Lynx*

1. Production of a master grating with sawtooth surface-relief profile that enables
   - Total absolute diffraction efficiency $> 40\,\%$
   - $\mathcal{R} \gtrapprox 5000$ in a Wolter-I telescope
2. Mass manufacture of grating replicas to populate modular grating arrays, which involves
   - Imprinting grating surface-relief molds
   - Coating each replica for soft x-ray reflectivity

Thermally-activated selective topography equilibration (TASTE)

- Couples GEBL with thermal reflow
  - ☐ Process development at Penn State Nanofab
- Eliminates dependence on crystal structure
  - ☐ Provides way for radially-ruled, blazed grating layout to be generated with EBL precision
  - $\Longrightarrow$ Enables high $\mathcal{R}$
- Meets efficiency requirement

Substrate-conformal imprint lithography (SCIL)

- Variant of NIL designed for high-throughput patterning over large areas
  - ☐ Equipment for low-volume SCIL recently installed at Penn State Nanofab
- Practical for *XGS* reflection grating production
  - ☐ Compensate blaze angle for resist shrinkage
- Comparable efficiency to UV-NIL replica

Acknowledgments

This research was supported by the NASA Space Technology Research Fellowship (NSTRF) under grant No. NNX16AP92H and used resources of

- The Advanced Light Source (ALS), which is a DOE Office of Science User Facility under contract No. DE-AC02-05CH11231
- The Nanofabrication Laboratory and the Materials Characterization Laboratory at the Penn State Materials Research Institute
- The Quattrone Nanofabrication Facility at the Singh Center for Nanotechnology, University of Pennsylvania
- Philips SCIL Nanoimprint Solutions in Eindhoven, Netherlands

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